

KSHITIJ MISHRA

AI Quality Analyst • LLM Evaluation • Prompt Engineer • AI Engineer
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PROFESSIONAL SUMMARY

Technical AI Engineer specializing in the design and architecture of scalable model evaluation systems and production pipelines. Expert in architecting automated evaluation frameworks for LLM and voice AI systems, optimizing retrieval-augmented generation (RAG) workflows, and engineering robust prompt suites. Proficient in Python, SQL, and deep learning frameworks (PyTorch), with a proven track record of reducing model hallucination rates and streamlining engineering lifecycles. Adept at bridging the gap between research prototypes and production-ready AI services through rigorous performance benchmarking and data-driven optimization.

CORE TECHNICAL SKILLS

AI & ML: LLM Evaluation, Prompt Testing, Output Validation, Hallucination Detection, Scikit-learn, PyTorch, Deep Learning

Programming: Python, SQL

Data Analysis: Pandas, NumPy, Exploratory Data Analysis, Trend Analysis, Data Cleaning, Kaggle Competitions

Voice AI: Whisper, Google STT, ElevenLabs, Coqui TTS

Tools: Power BI, Jira, Notion, GitHub, Structured Reporting

Domain: Cryptocurrency Market Data, Behavioral Pattern Analysis, Volatility Clustering

PROFESSIONAL EXPERIENCE

AI Quality Associate | Tavus AI

Jan 2024 – Present

- Architected and executed high-fidelity evaluation protocols for 500+ multimodal AI outputs monthly, ensuring structural and semantic integrity across live production streams.
- Engineered a multi-tier feedback architecture that reduced model hallucinations by 30% through the systematic injection of structured evaluation data into the research-to-engineering feedback loop.
- Optimized LLM retrieval and prompt-chaining workflows by implementing rigorous A/B testing and semantic validation, significantly increasing response consistency across 10+ production modules.
- Automated benchmarking pipelines using custom Python scripts to track drift and performance degradation, providing critical data for model retraining and fine-tuning cycles.
- Supported deployment readiness of 5+ AI features by validating performance across real-world multi-scenario test cases
- Collaborated with cross-functional teams to streamline evaluation pipelines, cutting review turnaround time by 20%

Relationship Manager | Urban Company

Aug 2022 – Sep 2023

- Analyzed customer behavior and partner performance data to identify service gaps, improving operational efficiency by 15%
- Maintained structured weekly performance reports used by operations leadership for data-driven decision-making
- Optimized partner onboarding workflow using data insights, reducing average onboarding time by 10%

Quality Engineer | Rashmi Rare Earth Ltd

Dec 2021 – May 2022

- Analyzed system performance data to drive process improvements and reduce throughput bottlenecks.
- Led technical root-cause analysis for process deviations, implementing engineering solutions to improve the system.

- Managed operational reporting pipelines, utilizing data metrics to optimize throughput and efficiency across production workflows.
- Developed and executed rigorous testing protocols for system validation, ensuring high standards of technical output and operational precision.

PROJECTS (Live demos available at kshitij.info)

AI Voice & Output Evaluation Framework

- Architected an end-to-end Python-based evaluation framework utilizing NLP metrics and custom scoring logic to programmatically validate LLM and TTS output quality.
- Developed automated validation gates within the CI/CD pipeline to catch logical inconsistencies and hallucinations before production deployment.
- Built scoring criteria and ran comparative test runs to benchmark prompt variants — improved output reliability by 25%

Tech: *Python, LLM Evaluation, Prompt Testing, Workflow Validation*

Crypto Market Analysis System

- Engineered a high-throughput data pipeline using Python, SQL, and Pandas to process and analyze massive cryptocurrency datasets for volatility clustering.
- Implemented time-series analysis modules and statistical modeling to identify behavioral signals, leveraging NumPy for optimized computational performance.
- Produced structured visual reports using Pandas and data visualization tools to support data-driven decision-making

Tech: *Python, Pandas, SQL, NumPy, Data Visualization*

AI Portfolio Agent — kshitij.info

- Developed a RAG-based AI agent using advanced prompt engineering and controlled logic to interface with structured technical documentation.
- Designed robust system prompts and verification loops that strictly enforce response constraints and eliminate unsupported model outputs.
- Applied prompt evaluation techniques to improve response accuracy and reduce unsupported outputs

Tech: *Python, Prompt Engineering, LLM Evaluation*

User Failure & Retention Dashboard

- Built a data-driven dashboard to analyze failure patterns and retention across AI-based workflows
- Tracked retry cycles, drop-off points, and cohort retention across 500+ users to identify key bottlenecks
- Generated actionable insights to highlight workflow friction and reduce user drop-off

Tech: *Python, Pandas, Power BI, Data Visualization*

EDUCATION

B.Tech — Electrical & Electronics Engineering

2017 – 2021

Dr. A.P.J Abdul Kalam Technical University | **75.6%**

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization — Andrew Ng, Coursera
- Google AdSense Approved Website — Developed a project during college
- Google AI Certification — 2021